

Metformin Clinical Guideline 2024

1. Indications and Clinical Usage

Metformin is indicated as the first-line pharmacotherapy for type 2 diabetes mellitus. It helps lower blood glucose levels primarily by decreasing hepatic glucose production and increasing insulin sensitivity in peripheral tissues.

2. Contraindications

Metformin is contraindicated in patients with severe renal impairment, defined as an estimated glomerular filtration rate (eGFR) below 30 mL/min/1.73 m². It should also be avoided in cases of acute metabolic acidosis, including diabetic ketoacidosis.

3. Adverse Reactions and Side Effects

Common adverse reactions include gastrointestinal side effects such as diarrhea, nausea, vomiting, flatulence, and abdominal discomfort. Rare but serious risks include lactic acidosis, especially in patients with renal dysfunction.